

RESEARCH CENTRE FOR INFORMATION SYSTEMS ENGINEERING (LIRIS)

An investigation into automating matching across touristic data

A case study on Airbnb and Tourism Flanders

Group report

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Abstract.
The rapid growth of the sharing economy, notably exemplified by Airbnb, has led to a significant increase in complex geospatial data. This data is vital for tourism analysis but challenging to integrate with official statistics. Accurately and efficiently matching these diverse geospatial datasets, particularly those from platforms like Inside Airbnb and official sources, is hindered by factors such as varying spatial accuracy and computational demands. These challenges resulted in knowledge gaps regarding the true scale of the short-term rental market, including accommodation capacity not captured in existing official datasets. This gap is of particular concern for regional stakeholders such as Tourism Flanders. In response, this research addresses the question: How can data matching techniques enhance the real-world validity and reliability of tourism statistics for the Flanders region? To address this, a multi-level data matching methodology was developed and applied, by using geospatial variables and unique identifiers. The methodology was evaluated by using a three-class classification framework: "Matched", "Missed" & "Inconclusive". It was subsequently concluded by estimating missed capacity and deriving the potential correction factor.

Keywords: Airbnb, Data matching, Geospatial.

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1 Introduction

Short term rental platforms like Airbnb, established in 2007, have grown into a major player in the global tourism industry. By 2024, it featured over 8 million active listings and 5 million hosts worldwide, demonstrating an impact on tourism that will be examined in this paper (Airbnb, 2024). Integrating Airbnb data with official tourism statistics is vital for understanding market dynamics, tracking regulatory compliance, and informing evidence-based policy decisions.

While the existing literature frequently employs Inside Airbnb data to estimate housing prices, tourism satisfaction, and tourist arrivals, the present study addresses critical knowledge gaps regarding the true scale of the short-term rental (STR) market. It is motivated by key questions posed by stakeholders such as Tourism Flanders. Tourism Flanders (2025) is the Flemish governmental tourism agency, dedicated to the promotion of Flanders as a high-quality tourist destination. This agency achieves its mission through data-driven strategic planning and a commitment to sustainability, accessibility, and family-friendliness. In response to Tourism Flanders' need for enhanced data on accommodation capacity, the primary objectives set out by Tourism Flanders are summarised below:

1) Find the locations missed by Tourism Flanders.
2) Use that to calculate the missing capacity.
3) Develop a factor to adjust Tourism Flanders overnight stay statistic.

These objectives are a guideline for answering the research question: How can data matching techniques enhance the real-world validity and reliability of tourism statistics for the Flanders region? The motivation for this question lies in bridging the gap between the listings available on the Airbnb platform and corresponding records in the official register for Antwerp and Ghent. By providing clearly structured steps, this approach aims to facilitate more efficient data matching and potentially reduce the time required for the data matching process. Hence, achieving these objectives is crucial for obtaining a more accurate picture of tourism activity and its economic impact.

The challenge of measuring the full effect of short-term rentals on the Airbnb platform, is often due to the lack of comprehensive or readily available data for integration with official sources. This gives rise to the concept of "unknown capacity" that the accommodation supply does not capture in official statistics. Therefore, to address Tourism Flanders' need for enhanced data, this thesis aims to develop a correction factor derived from the estimated unknown capacity. This factor could then be applied to adjust official overnight stay statistics, providing a more accurate measure of tourism activity.

This more accurate measure would enhance the accuracy and utility for regional planning, resource allocation, and policy formulation.

The rapid growth of the sharing economy, particularly in the accommodation sector, has highlighted the challenge of collecting and harmonising data for this innovative business model, as conventional tourism data sources are often insufficient (Ferreri & Sanyal, 2018; Niestadt, 2024). Researchers and stakeholders seek to understand this market and access information through various channels, including direct platform provision, commercial datasets (e.g., AirDNA), and web-scraped open-source initiatives like Inside Airbnb. Web scraping is defined as "the activity of taking information from a website or computer screen and putting it into an ordered document on a computer" (Web Scraping, 2025). Relying on these alternative sources, however, often presents significant data quality and reliability concerns (Prentice & Pawlicz, 2023; Alsudais, 2021). Issues include data heterogeneity and semantic inconsistencies in attributes, incomplete records leading to potential undercounting and reluctance from platforms to share detailed data necessary for verification and regulation (Cox & Haar, 2020; Prentice & Pawlicz, 2023). Crucial for geospatial analysis and accurate record linkage is that these datasets often exhibit different levels of spatial accuracy, including documented instances of intentional coordinate deviation for privacy purposes in open-source data.

Accurately and efficiently identifying and matching corresponding listings across diverse geospatial datasets with varying structures and semantic definitions, however, presents significant challenges. First, data heterogeneity and semantic inconsistencies, such as differences in classifications or definitions for the same concepts of geographic features, complicate direct comparisons and can lead to ambiguity in matches (Riedl et al., 2006). Second, these datasets often exhibit differing levels of spatial accuracy, including intentional coordinate deviation for privacy purposes common in open-source data, making proximity-based matching unreliable without specific adjustments (Christen, 2012). Third and finally, beyond these data quality and representational issues, there is a substantial computational demand associated with comparing potentially large datasets, particularly when large amounts of comparisons are required (Christen, 2012; Lei, 2021). These complexities are particularly pronounced when attempting to match data from sources with different data collection methods and purposes, such as open-source platform scraping versus static official registers. Effectively handling these complexities is crucial because inaccurate or inefficient matching can lead to flawed downstream analyses, incomplete datasets and ultimately, misinformed decisions in practical applications.

To overcome the inherent challenges, this research proposes and validates a four-phase data matching methodology to answer the first objective by Tourism Flanders following up with two more phases of the method to answer the last two. This structured approach systematically refines the matching process

across distinct analytical levels to enhance accuracy and efficiency in integrating heterogeneous geospatial datasets with spatial accuracy. The first level utilises IDs and corresponding web addresses for initial matches between the Inside Airbnb and official data. Building on this, the second level leverages geospatial variables to clarify and refine potential matches, specifically accounting for spatial uncertainty. The third level employs a descriptive variable match and a search function to further confirm missed locations through attribute comparison. Subsequently, the fourth level applies the outcomes identifying the matches and the missed listings from the preceding levels to derive an approximation ratio for estimating the total missing locations. The penultimate part of the methodology estimates the overall unknown capacity relative to the total market size. Finally, this is used to derive a factor for adjusting the overnight stay statistics of Tourism Flanders.

The rest of the paper is structured as follows: The second section provides a general overview of the major scholarly literature, focusing on research trends in estimating overnight stays, the application of Airbnb data in studies, and existing geospatial data matching methodologies. The third section describes the data sources utilised in this work, elaborating on the innovative strategy, explaining how it was structured into four distinct phases, each applying a different matching approach. The fourth section presents the results obtained from this process. Finally, the last section concludes with a discussion of the key findings and their implications for regional planning.

2 Literature Review

This section provides a review of relevant academic research on the short-term rentals (STRs) and their impact on the tourism market. To gain insights on an approach to answer the three objectives set by Tourism Flanders outlined in the introduction. It begins by outlining the potential effects of short-term rentals. The review then explores the challenges associated with data collection in the STR sector. It continues with discussing methods for estimating the number of overnight stays. Finally, geospatial data matching methodologies are evaluated, focusing on linking diverse spatial datasets, particularly from platforms like Inside Airbnb and official sources, to pinpoint the research gap.

2.1 Economic and Regulatory Challenges of STR

The proliferation of STRs, exemplified by Airbnb, has generated significant and complex impacts across various sectors, presenting challenges for regulators seeking to balance economic benefits with potential downsides. Research highlights considerable disruption to the traditional hospitality industry, particularly impacting lower-end hotels facing direct price competition and revenue

loss (Chang & Sokol, 2020; Zervas et al., 2017). Beyond hospitality, concerns extend to potential tax avoidance by hosts, straining public revenue and negative pressure on local housing markets through the conversion of residential stock, potentially reducing long-term supply and increasing rental prices (Bibler et al., 2021; Ključnikov et al., 2018; Schäfer & Braun, 2016). Furthermore, crucial for regulation and monitoring, a significant tension exists between broad digital service laws and the localised nature of STR impacts. This is exacerbated by the reluctance of platforms to share granular, georeferenced data necessary for effective enforcement of local rules (Ferreri & Sanyal, 2018; Niestadt, 2024).

2.2 Challenges & Limitations in STR Data Collection

A significant challenge in the collection and harmonisation of data for the sharing of STRs has raised data reliability concerns and issues with the level of detail (Ferreri & Sanyal, 2018; Niestadt, 2024). The STR business model utilised by digital technology companies such as Airbnb, generate vast amounts of data but resist sharing it, a practice that complicates efforts to accurately assess market dynamics and regulatory compliance. Unlike traditional sectors, for example hotels and Bed & Breakfasts (B&Bs), which often contribute to more transparent or accessible data pools through established reporting, platforms like Airbnb operate with a degree of data opacity that makes their data difficult to capture through conventional means. The necessity for alternative methodological approaches to data collection and analysis has been highlighted in several key studies. Researchers can access information on Airbnb activity through the three primary channels: direct platform provision, commercial datasets (e.g., AirDNA) and open-source data initiatives based on web scraping (e.g., InsideAirbnb). Exemplifying the use of commercial data, Seiler et al. (2023) investigated the causal effect of Airbnb bans on long-term rental markets, using Irvine California as a case study. They combined data from AirDNA with data from CoreLogic (CoreLogic, 2025). Both of which adding relevant information, with AirDNA measuring Airbnb activity through metrics like active listings, occupancy rates, and reservation sums and CoreLogic providing long term rental market information, including geospatial data and Multiple Listing Service (MLS) listings. Utilising a Difference-in-Difference (DID) methodology, their research aimed to estimate how Airbnb activity, and specifically its cessation due to bans, impacts long-term rental prices and supply, demonstrating the utility of integrating comprehensive commercial datasets for such analyses.

While research like Seiler et al. (2023) utilises commercial datasets, obtaining accurate data directly from platforms like Airbnb for regulatory purposes presents significant challenges. Despite pressure from governments in cities such as Amsterdam, Barcelona, New York, and Paris to disclose detailed listing

data, compliance has been problematic. For example, an agreement with Barcelona yielded data where 60-70% of addresses were missing or incorrect, severely hampering efforts to verify listings against the city's registration system (Cox & Haar, 2020).

Nevertheless, publicly open sources based on web scraping could be a solution for the data and method transparency. Inside Airbnb as one of the most well-known open-source platforms, has helped to provide insights for urban planning regulators and academic researchers. Gutiérrez et al. (2017) exemplify the spatial analysis application, utilising Inside Airbnb data to map and compare the spatial distribution of Airbnb listings against traditional hotels in Barcelona. While the paper does not explicitly address the problem of inaccurate location data associated with Inside Airbnb, their study assumes a sufficient level of locational accuracy for their macro-level spatial analysis. By employing geographical mapping in GIS software and spatial statistical techniques like Exploratory Spatial Data Analysis, including Mo-ran's I and LISA statistics, the study effectively visualised and quantified the distinct spatial distributions of Airbnb and hotels across Barcelona. As another example, Balcıoğlu (2024) used Inside Airbnb data in the region of Istanbul to identify the features for guest's experiences. The study utilised geographical mapping as a key component. He employed the variable with a geographical feature and R to map accommodation units and visualised these locations in relation to Google Maps data. Balcıoğlu aimed to verify the reviews were relevant to the actual locations of the Airbnb Listings by correlating this mapped location with the thematic content of reviews associated with them. This geographical framework then enabled the analysis of review content based on location, allowing for the examination of spatial distribution of review themes and the validation of their relevance to Istanbul.

The methods described above used by Gutiérrez et al. (2017) and Balcıoğlu (2024) offer some insight into how Inside Airbnb data may be used to create a map of Airbnbs. However, Alsudais (2021) highlighted in his review two critical issues arising with this usage of Inside Airbnb data: incorrect data and reproducibility problems. Specifically, it appears that reviews for Airbnb "experiences" were mis-linked to the listings for "places". Moreover, the paper mentioned that the reviews for activities in one location were attached to accommodation listings in another. These concerns underscore the broader point that relying on unverified open datasets like Inside Airbnb presents inherent risks and quality issues that would limit the accuracy and transparency for future research. Following the findings by Alsudais, other research such as Prentice and Pawlicz (2023) and subsequent scholarly work (La et al. 2021, Wang et al. 2024, Zhang et al. 2020), examined the limitations of Airbnb data obtained through web scraping (Inside Airbnb for the sharing economy sector). The main issue for collecting the sharing economy rentals, as mentioned in their paper, is that the STR market is not recorded in the official statistics system. Platforms

such as Airbnb and other STRs are reluctant to share data due to the privacy, competition, and regulatory factors (Prentice & Pawlicz, 2023). As mentioned by Prentice and Pawlicz (2023) limitations for these major short-term rental market data sources might include inactive listings (where some listings are fake, outdated or overpriced). These would result in higher average prices and might imply a lower occupancy rate. On top of this, the biased geographical locations in-crease the difficulties for their academic research paper, this is also an ever-present issue in our study. Due to a privacy issue, the geographical locations obtained from the Inside Airbnb platform has an inaccuracy of up to 150 metres. This is a vital element for estimating price, the distance between properties and for geographically data matching when address checking with formal data provided from government statistics (Prentice & Pawlics, 2023).

Furthermore, Prentice and Pawlicz (2023) has stated that missing listings due to unavailability of listings, and exclusion due to reservation query settings could result in a potential undercount of around 20% in the research. Their review revealed that some papers did not realise the limitations, or the quality of data sources obtained from self-made bots, InsideAirbnb or AirDNA.

2.3 Geospatial Data Matching

The first objective from Tourism Flanders concerns discovering missed locations. This task is geospatial in nature due to the presence of geospatial variables in the dataset. A significant challenge in integrating geospatial datasets arise from matching corresponding features, a term Lei (2021) coined "geospatial data conflation", which he defines as "the process of combining two datasets to create a better one'' (p. 1). Initial analysis often relies on seemingly straightforward coordinate matching (latitude and longitude). This frequently reveals the necessity for deeper investigation into conflation methods. Lei (2021) outlines numerous sources of matching errors, such as human-caused error specifically when dealing with property data. The variability in property size and distance means that assigning a single representative geospatial point involves considerable discretion, leading to potential inconsistencies between datasets even for the same underlying address. This issue appeared in datasets like Inside Airbnb, which intentionally introduce coordinate inaccuracies of up to 150m for privacy protection, further complicating direct point matching and motivating the exploration of optimal matching methods.

The study by Lei (2021), discusses various conflation methods, including optimisation techniques aimed at achieving high precision. Lei (2021) also highlights a common drawback of such optimal methods: high computational load. To mitigate this the "divide-and-conquer strategy" approach has been proposed, reportedly improving processing time while maintaining matching precision. Further contributing to this area, Zhang et al. (2020) ´s paper examines methods that attempt to combine non-optimal and optimal techniques for

geospatial data matching. Zhang et al. (2020), introduces a "georeferenced-graph model", adapting upon a polygonal approach which represents area features (like building footprints) creating an outline of the surface area of the building on a map and utilising it to measure the distances between each building. With the challenge of recognising the ambiguity inherent in defining geographic points of a building highlighted by Lei (2021), Zhang et al. (2020) tackled this problem by using centroids, a concept created by Archimedes (287-212 B.C.). Defined by Apostol et al. (2000) as "a special case of a more general concept called centre of mass", following Archimedes as a geometric centre, centroids provide a standardised point within each polygon for establishing spatial relationships. Zhang et al. (2020) utilise a GIS (Geographic Information System), described as systems for managing and analysing positional data to derive these footprints and centroids. They can then map relationships onto these points to create georeferenced graphs, facilitating optimised investigation of spatial connections.

The central contribution of Zhang et al. (2020) relevant to this research lies in their methodology designed to accurately match identical real-world entities across different datasets by considering both area and entity scales. One of the limits of Zhang et al. (2020), however, was that their results are based on a single dataset. Their model faces challenges when attempting to match across datasets that lack inherent correspondence, highlighting an area for future research.

2.4 Estimating the Overnight Stay Statistic

The second objective concerning the missed capacity can be answered based off a variable in the dataset once the first objective is achieved. The third objective to adapt their overnight stay statistic, however, presents a greater challenge so prompts further research. Overnight stays estimation is a key analytical task within tourism and hospitality research, which involves developing reliable methods to achieve it. When estimating overnight stays, variables that have sociodemographic and trip-related characteristics, supply-based factors and knowledge of destinations are often used (Barros & Machado, 2010; Boto-García et al., 2018; Rodríguez et al., 2018).

The most common statistical models that are applied when investigating the number of overnight stays are survival models (duration models), Ordinary Least Squares (OLS) regression and the Heckman model. Gokovali et al. (2007) employed survival analysis, which is a technique analysing the time elapsed between a tourist's arrival and departure. The authors' analysis was done on a questionnaire survey conducted in the summer of 2005 in Bodrum, Turkey. The results have indicated that sociodemographic characteristics include nationality, education, and income. Trip experiences and daily spending are the most impactful variables that determine the duration of the stay. Despite the subtle

differences in method, Martinez-Garcia and Raya (2008) found a strong positive effect of age on the overnight stay. Tourists over 50 years old tend to stay longer, contrasting to Gokovalie et al. (2007) who found that there was no significant effect for different age ranges in their model.

Employing data from the summer of 2007, Thrane and Farstad (2012), estimate an OLS regression model. While acknowledging the prior use of survival analysis models in similar studies (eg., Gokovalie et al., 2007; Martinez-Garcia and Rayal., 2008), Thrane (2012) argues its suitability for the context for estimating overnight stays. The author aims to determine the extent to which nationality and other independent variables influence the duration of visitor stays to inform tourism policy. In the study, nationality has accounted for a portion of 65% of the overall explanatory power of the model. As mentioned in another study by Thrane & Farstad (2012) the other variables such as age, expenditure per day, and number of places visited have a significant impact on the length of stay.

The study of Rodríguez et al. (2018) was the first employing the Heckman model in this field of study to correct for sample selection bias. The authors used it to investigate length of stay, which separates the analysis into two parts corresponding to two dependent variables. In the first selection equation there was a binary variable that captured whether a visitor stayed overnight or not. The second, modelling the outcome process, had a discrete numerical variable representing number of nights the tourists stayed. However, their findings did not indicate notable sample selection bias where the inverse Mills ratio was statistically insignificant. It suggests that a simpler truncated regression model would likely suffice to provide consistent estimates for the number of overnight stays.

Machine learning (ML) has been explored for tourism demand forecasting as an alternative to statistical models for tackling the estimation of overnight stays. For example, the study by Alegre and Pou (2006) explores the binary logit models (staying 7 days or less versus staying more than 7 days) to examine the length of stay over the microeconomic determinants in the region of the Balearic Islands. Using data from the Tourist Expenditure Survey (TES) from 1993 to 2003, Alegre and Pou (2006) discovered that tourist stays are significantly influenced by their sociodemographic background, trip characteristics and sensitivity to price. Another example of ML is by Jackman and Naitram (2023) who were among the first to employ regression tree analysis for duration of overnight stays, with data obtained from the Barbados Immigration Department from 2004 to 2013. Significantly, they demonstrated that accommodation type was the most influential feature, indicating that hotel stays (average of 9.9 days) were shorter compared to stays in other accommodation categories (average of 16.6 days). Expanded upon by Singgalen (2024), the study offers a robust approach for predicting hotel guest length of stay by integrating the Random Forest Regression model. The utilised variables are sociodemographic

(country, guest type etc.) and trip related characteristics (room type, rating etc.). Their approach by using ensemble learning with numerous decision trees is capable of handling both categorical and continuous variables. The key findings of Singgalen (2024) were that there was only an error of one night between the predicted and actual results when using the evaluation metrics of Mean Squared Error (at 1.06) and Root Mean Squared Error (at 1.41). The most significant variables that were identified in the analysis are country, guest type, room type and ratings.

Antolini and Cesarini (2023) provide another example by leveraging microdata from the Italian National Institute of Statistics that focused on predicting the length of stay for domestic tourists in Italy by employing Random Forest, Gradient Boosting, and Multi-Layer Perceptron (MLP). their results indicate that the performance of the machine learning models on the test set was strong, evidenced by high positive linear correlations between predicted and actual values, with high correlation coefficients reaching 0.93 (Random Forest), 0.98 (Gradient Boosting), and 0.99 (MLP). Such high performance suggests these models, particularly MLP and Gradient Boosting in this case, are well-suited for modelling the nuances influencing domestic stay length based on detailed microdata.

A notable drawback for estimating number of overnight stays with the variables from the Inside Airbnb is that, to our knowledge the available variables are primarily focused on listing features. They may lack crucial information about guest behaviour or motivations that significantly influence stay duration. Factors like trip purpose, travel party size and visitor demographics which are strong predictors of length of stay as mentioned in the preceding literature reviewed, are generally absent.

3 Problem Statement

Six important points were identified in the literature review. First, there is a clear issue with the lack of regulation of STR websites, which has consequences for businesses and government. The issue stems from the fact that platforms do not want to share detailed information about their listings. Second, data for STR websites is difficult to collect. One solution is to use open-source data, such as Inside Airbnb. This presents its own challenges and problems which are not considered in all scientific work. Third, there are problems matching geospatial data specifically with coordinates originating from different building footprints and observers. Fourth, data matching over these coordinates can cause a high computational load which can be tackled by splitting the task into smaller tasks. Fifth, there are also issues with applying previously designed methodologies for data matching to real world entities over different datasets. Sixth, there are

different methods for estimating overnight stays with or without the use of machine learning.

The literature review revealed the lack of a clear methodology that considers the inconsistencies in the data and the problems encountered when using STR data and which, despite these limitations, provides a reliable match for tracking the amount of STRs currently on the market. The study will use the open-source dataset Inside Airbnb to tackle the issues in STR data collection while acting upon its limitations. Tourism Flanders provided the objective of improving their data and attempting to track these STRs in their region. They tried to do this manually, but it proved infeasible due to time constraints and human error. Showing the importance of automated data matching techniques and leading to this study's research question: How can data matching techniques enhance the real-world validity and reliability of tourism statistics for the Flanders region?

4 Methodology

The methodology aimed to answer the research question via achieving the three objectives set by Tourism Flanders, which resulted in an estimate of the missed overnight stays. To accomplish this task, the methodology was broken down into three steps. The first step was pre-processing, an often underestimated but essential phase that typically accounts for over 50% of the total effort (Pyle, 1999). The second step was data matching, during which entities are compared based on various parameters and descriptive variables to identify properties that may have been over-looked by Tourism Flanders. Finally, missed capacity was calculated leading to the calculation of a factor that was used to adapt Tourism Flanders overnight stay statistic.

4.1 The Data

Based on insights from the literature review, which highlighted the advantages of using open-source data, this study selected the Inside Airbnb dataset. It was compared to open-source data provided by Tourism Flanders. The study utilised these two primary datasets to investigate the optimisation data matching methods in the context of short-term rentals, and specifically focused on Ghent and Antwerp, Belgium.

Inside Airbnb Datasets

The Inside Airbnb's data collector web scraped the Airbnb data directly from the Airbnb platform. It was initially comprised of 4,100 listings: 2,757 in Antwerp and 1,343 in Ghent. Each listing includes a unique ID, which serves as the primary key within the dataset and corresponds to the identifier used in

Airbnb web addresses. This ID was crucial in the initial indexing and matching phase, as detailed in the data matching section. A direct link to each listing, labelled as “listing_url”, is also included, providing immediate access to the Airbnb website. The dataset includes geospatial coordinates (latitude and longitude), though these are intentionally imprecise, with deviations of up to 150 meters, introduced for privacy reasons. This limitation was directly addressed during the geospatial matching phase of the project. The dataset also includes a variable named “accommodates” which represents the max capacity an accommodation can host at a given time. The additional variables present in the dataset, but not used in this thesis, include review data (e.g., last review date, number reviews per month), logistical details (price, availability throughout the year, room type, etc), and host related information (e.g., host name, host ID, total number of listings per host).

Tourism Flanders Dataset

The Tourism Flanders dataset contains 30,114 entries representing various accommodations across Flanders, including listings identified as Airbnb rentals in Ghent and Antwerp. Each entry has a unique “business_product_id”, which serves as the dataset’s primary key. The dataset includes detailed geospatial information namely city name, street, postal code, house number and box number, which together form the full address of each property as well as latitude and longitude. While these are formatted similarly to those in the Inside Airbnb dataset, they are more precise, as they do not include intentional inaccuracies. These coordinates were used in the geospatial matching process. In addition to geolocation data, “web_address”, a URL linking to the property's official Airbnb page and “max_capacity”, indicating the maximum number of guests the accommodation can host, were two key variables used in the matching process. A variable that was used while adapting the overnight stay statistic was “discriminator” this contains a brief description on the type of properties listed. For example, “BED_AND_BREAKFAST”, “GENERIC_ROOMS” & “HOLIDAY_COTTAGE”. Other variables present in the dataset but not used in this thesis include the accommodation’s name and the host’s contact details (phone number and email address). A main challenge of this research is to link properties in the Tourism Flanders dataset to their corresponding listings in the Inside Airbnb dataset using the selected variables.

4.2 Pre – processing

To enable accurate data matching, pre-processing was applied to the Tourism Flanders and Inside Airbnb datasets by filtering relevant entries, cleaning and standardising URLs, removing inactive listings, and extracting room IDs using Excel tools and a custom web scraping script.

The pre-processing began with the Tourism Flanders dataset, which was filtered to include only listings located in Ghent and Antwerp that were associated with Airbnb. This was done by selecting entries containing both an Airbnb web address and the appropriate city name. Duplicate entries were also removed as part of the initial data cleaning. The next step addressed issues with the website field, many links were either incomplete or incorrectly formatted, with some missing the Airbnb room ID entirely. To standardize these addresses and improve the accuracy of our indexing, web scraping was implemented. A 5-second timeout was then applied to check whether each link was accessible as seen in the Fixing web addresses subprocess in Fig. 1. If a page responded, the HTML content was converted to plain text and subsequently pattern matching was used to extract the valid room ID in the format used by Inside Airbnb.

For the Inside Airbnb dataset, the separate Ghent and Antwerp files were first combined into a single dataset. Before merging, a “City” column was added to each file to preserve the geographic origin of each listing. After appending, quality checks were performed to identify and remove duplicate entries or listings with missing location data. Although the website links in this dataset were already complete and correctly formatted, some properties were no longer available on Airbnb. To address this, a 5-second timeout check was applied to identify and remove these inactive listings. This process was comparable to the Fixing web addresses sub process but with a different outcome as shown in Fig. 1. Finally, the Airbnb dataset was consolidated into the same document as the Tourism Flanders dataset.

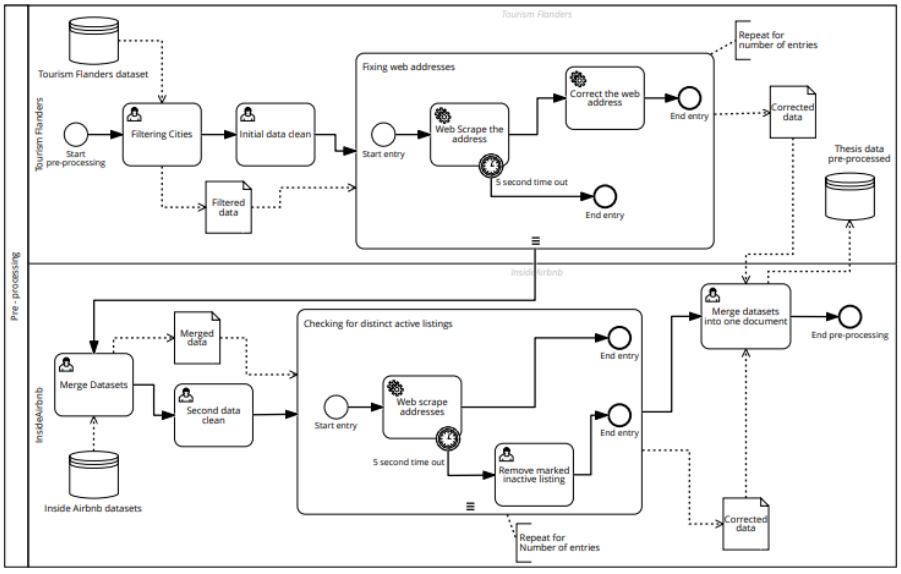


Fig. 1. Pre-processing BPMN, outlining the general pre-process, including filter, clean, fixing web addresses, checking for distinct active listings and merge data.

4.3 Data Matching

This section outlines the approach used to perform data matching on the pre-processed datasets. The matching task can be viewed as a three-class classification problem, with the categories: “Matched by Tourism Flanders”, “Missed by Tourism Flanders”, and “Inconclusive” (Christen, 2012). The objective was to minimise the list of inconclusive cases. The process was broken down into four phases. The phases were done to break up the task and reduce computational load and are represented in Fig. 2. It began with matching the ID in the website links with the primary key from Inside Airbnb. In the next two phases, additional descriptive variables were used to narrow down further. The overlapping variables selected from each dataset were specifically geospatial coordinates and the maximum capacity of the property. The final phase involves applying an approximation to complete the task.

Phase 1: Indexing

To ensure computational efficiency in geospatial data matching, the goal was to exclude records that did not require comparison. Indexing, the second stage of data matching as outlined by Christen (2012), typically reduces the number of comparisons by filtering out unlikely matches. However, due to the limited number of shared variables between the two datasets, a different approach was adopted. Instead of identifying unlikely matches, the focus was on confirming exact ones. The primary key in the Inside Airbnb dataset is a unique identifier that is also used by Airbnb in their website URLs. As mentioned earlier, the Tourism Flanders dataset contains these web links. Matching was carried out by extracting the listing ID embedded in the URLs and comparing it with the Airbnb dataset’s primary key. If the IDs were identical, the listing was classified as “Matched by Tourism Flanders”.

Phase 2: Geospatial Matching

The second stage of the process was to match based on the distance between the geospatial data points. Due to geospatial inaccuracies in both the Inside Airbnb dataset and general geolocation data, a distance threshold of 172 metres was applied. This consisted of the 150-metre inaccuracy noted in the Inside Airbnb data dictionary, plus a margin of 11 metres on each side, derived from the square root of the average housing surface area in Belgium (Evans & Hartwich, 2005). This was done to address the problem raised in the literature review of potentially having different observation points to report the coordinates. Decreasing this threshold would risk excluding real matches. To determine whether two points fall within this threshold, the Haversine formula was

applied. This formula (outlined below) was adapted from a paper by Azdy & Darnis (2020).

$$\text{distance} = 2 \times RE \times \arcsin\left(\sqrt{\sin^2\left(\frac{\theta_2-\theta_1}{2}\right) + \cos(\theta_2) \times \cos(\theta_1) \times \sin^2\left(\frac{\phi_2-\phi_1}{2}\right)}\right) \quad (1)$$

In this formula, *RE* = Radius of the Earth (6371 km) (Ward & Ward, 2025).θ_1 and ϕ_1 represent the latitude and longitude of the Inside Airbnb dataset, and θ_2 and ϕ_2 refer to those of the Tourism Flanders dataset. The distance represents the separation between the two points. The Haversine function was chosen for its accuracy over short distances and its computational simplicity (Azdy & Darnis 2020). The geospatial matching analysis was conducted in Python using a custom function provided in Appendix 1 the function iterates over both datasets represented by a sequential multi-instance sub process embedded inside of another in Fig. 2. For each Tourism Flanders entry, it calculated the Haversine distance to every Inside Airbnb entity and includes those within 172 metres in the result. The output is a dictionary where each Tourism Flanders entry is a key mapped to a list of matching Inside Airbnb entities.

Phase 3: Capacity

This phase employed the dictionary computed in the previous phase, along with the "max_capacity" variable from Tourism Flanders and the matching accommodates variable from Inside Airbnb which both represent the capacity of the properties. Each key in the dictionary of matched geospatial locations was examined to retrieve the listing's maximum capacity. For each key, the associated values are also assessed by comparing their "accommodates" values to the key's "max_capacity". If a value's capacity does not match that of the key, it is confirmed that there is no match and thus removed from the "inconclusive" category and classified as missed by Tourism Flanders.

Phase 4: Approximation

To start the fourth phase, the ratio of Tourism Flanders entities not included in the Inside Airbnb dataset was calculated based on the confirmed matches and confirmed non-matches. Specifically, this involved dividing the number of unmatched Tourism Flanders entries by the total number of confirmed entries excluding the unknowns. Subtracting this ratio from one gave us the "Approximation ratio" which was used to estimate the proportion of Tourism Flanders entities found in the Inside Airbnb dataset. This proportion was then applied to the remaining Tourism Flanders entries to estimate how many of them are likely present in the Inside Airbnb dataset. This estimated number was subtracted from the total number of unmatched Inside Airbnb entries, providing an

approximation of how many listings may have been missed by Tourism Flanders. The final output of confirmed matches and missed by Tourism Flanders is represented by a signal end event in Fig. 2.

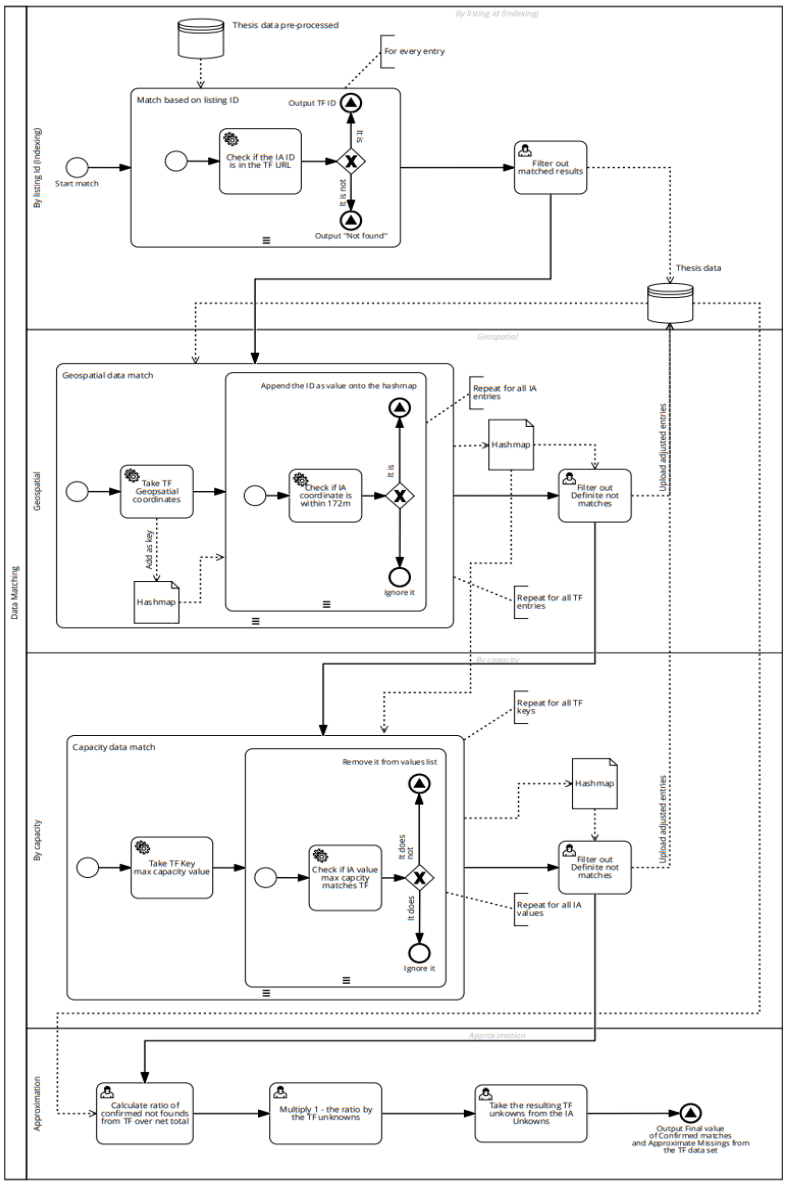


Fig. 2. Data Matching BPMN, illustrating the general data matching process, including Match based on listing ID, Geospatial data match, Capacity data match and Approximation. IA stands for Inside Airbnb and TF stands for Tourism Flanders.

4.4 Testing

The testing approach assumes that there is completely accurate data. However, this assumption may not hold in all cases. As the STR market is inherently volatile which means that key variables may be outdated and the use of scraped data introduces uncertainty in the verification of listing authenticity. These issues with this assumption will be further elaborated on in the limitation section. Under this assumption, the indexing phase gives confirmed matches and therefore does not require further testing. The results of this phase, however, can be used to evaluate the geospatial matching process. Specifically, by examining the recall of the geospatial matching method. That is, whether these confirmed matches from the indexing phase are correctly retained and not incorrectly excluded (Wang & Zheng, 2013). This helps ensure that the chosen distance threshold does not inadvertently exclude valid matches. The third phase again does not require additional testing under the correct data assumption. The approximation phase assumes that relationship between the matched pairs of previous stages is representative of the entire dataset. However, this assumption cannot be tested without knowing beforehand the exact matches of the entire dataset. This is a task that would require a physical visit to each location to verify the match due to the limited variables and is beyond the scope of this paper.

4.5 Missed Capacity

This section will elaborate upon the method used to achieve the second objective from Tourism Flanders. To calculate the missed capacity, the calculation begins by identifying the locations confirmed to be missed from Inside Airbnb and adds up their values for max capacity from the "accommodates" variable. Based on this, the average of the max capacity was computed and multiplied by the 362 approximate missed, adding it to the total. From these results a factor was developed to help estimate the overnight stays by defining the ratio of missed capacity plus the original capacity over the original capacity:

$$\text{Factor} = \frac{\text{Missed capacity} + \text{Original capcity}}{\text{Original capacity}} \tag{2}$$

4.6 Overnight Stay Statistic

To estimate the number of missed overnight stays, the calculation began by summing up individual overnight stay statistics from 2024 as provided by Tourism Flanders. This advantage was not present in the papers in section 2.4 of the literature review because of this advantage the statistic can be adapted from the original instead of entirely estimated. This total was then filtered according to

relevant accommodation types. Specifically, this concerns the categories marked under the “discriminator” variable in the Tourism Flanders dataset. This variable provides a brief description of each accommodation type. In this case: “BED_AND_BREAKFAST”,“GENERIC_ROOMS” & “HOLIDAY_COTTAGE” were used as those are the only ones that were in the remaining entries of the Airbnb’s after the pre-processing step. Before the final step, the proportion of each selected category represented by the filtered Airbnb data was calculated relative to the full dataset. This proportion was then applied to the total overnight stays reported in each category to estimate an equivalent “original overnight stay” figure for Airbnb accommodations. Finally, the adjusted estimate for missed Airbnb overnight stays was obtained by multiplying these figures by the previously calculated proportion, resulting in an approximate value for missed stays attributable to untracked Airbnb listings.

$$\text{Adapted over night stay statistic} = \text{Factor} \times \text{Original statistic} \quad (3)$$

5 Results

This section first provides a detailed breakdown of the matches and confirmed non-matches identified during each phase of the data matching process. These results are illustrated using visualisations from the city of Ghent. Due to space constraints, visualisations for Antwerp are included in Appendix 1. Second, following the data matching analysis, the section discusses the results of the calculations of missed capacity. Third and finally, it examines the adjusted overnight stay statistics. In all visualisations, Inside Airbnb data is represented in orange, while Tourism Flanders data is represented in blue.

5.1 Data Matching

Originally, the Tourism Flanders dataset contained 30,114 entries, and the Inside Airbnb dataset contained 4,100 entries. Following pre-processing, which involved filtering by city, removing duplicate or inactive listings, excluding entries with missing location data, and discarding those with invalid or incomplete Airbnb links. these were reduced to 468 and 3412 entries, respectively. Fig. 3 provides a visualisation of the location data of the entries from both datasets in Ghent.

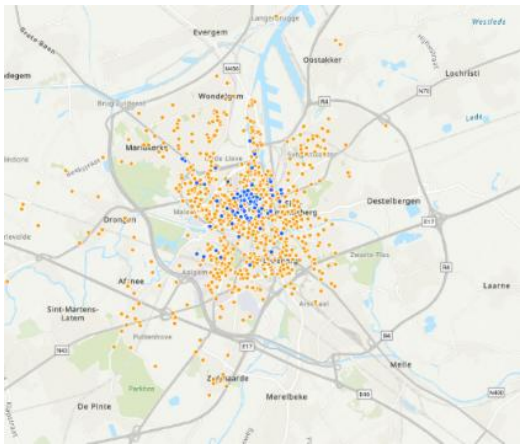


Fig. 3. Original pre-processed data entries from Inside Airbnb and Tourism Flanders.

During the indexing phase, 286 confirmed matching entries were identified between the two datasets. For the Tourism Flanders dataset, this reduced the number of unknown entries by 61.11%, due to its smaller total size compared to the Inside Airbnb dataset. Fig. 4 provides an overview of the 286 matched entries, while the remaining unmatched or unknown entries are shown in Fig. 5.

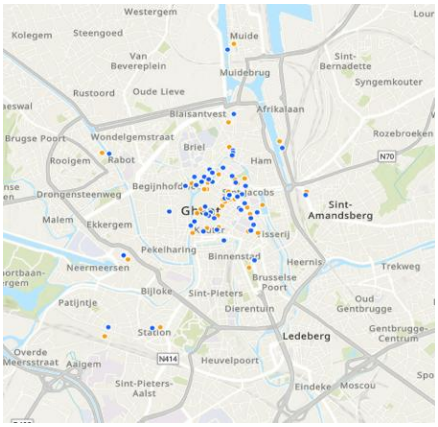


Fig.4. Indexing Phase: Visualisation of the matched entries.

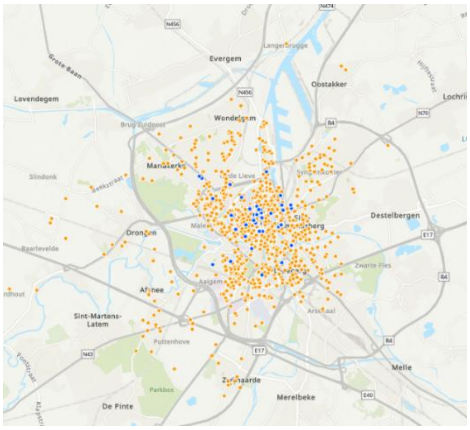


Fig.5. Indexing phase: Visualisation of the unmatched and unknown entries.

After the geospatial matching, an interesting phenomenon occurred, six confirmed non matches were extracted from the Tourism Flanders data set. This can likely be attributed to the volatile nature of Airbnb data. The Inside Airbnb Dataset had 2,105 confirmed non matches. This reduced the ratio of unknowns for Inside Airbnb by 67.34%. Fig. 6 shows the confirmed non matches, and Fig. 7 shows the still unknown data points.

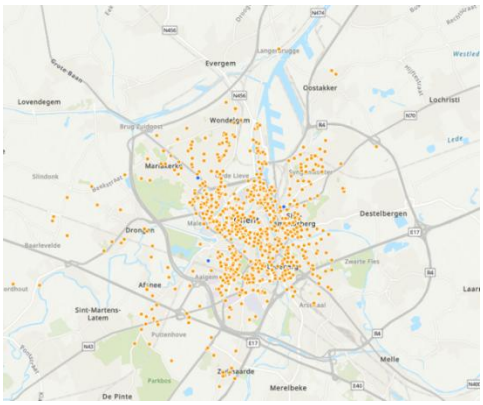


Fig. 6. Geospatial matching phase: Visualisation of the confirmed non matches.

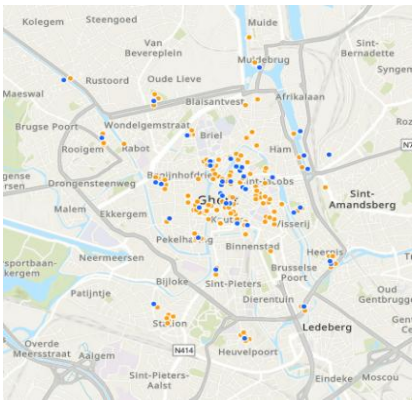


Fig. 7. Geospatial matching phase: Visualisation of unknown data points.

After the capacity check phase, 47 confirmed non-matches were identified from the Tourism Flanders dataset and 551 from the Inside Airbnb dataset. These confirmed non-matches are visualised in Figure 8, while the still unknown entries are shown in Fig. 9.

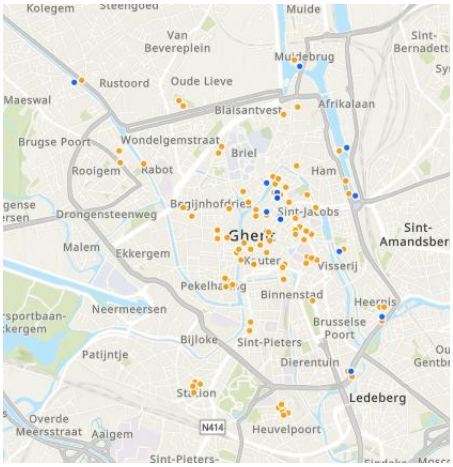


Fig. 8. Capacity check phase: confirmed non-matches.

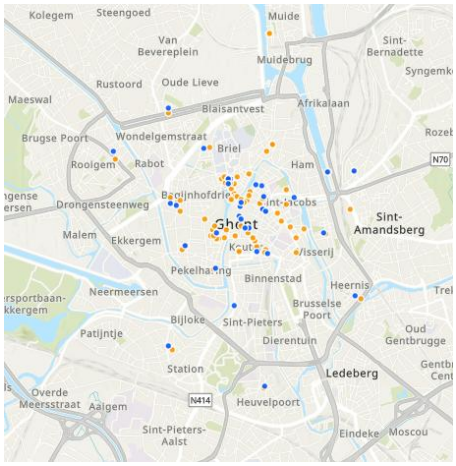
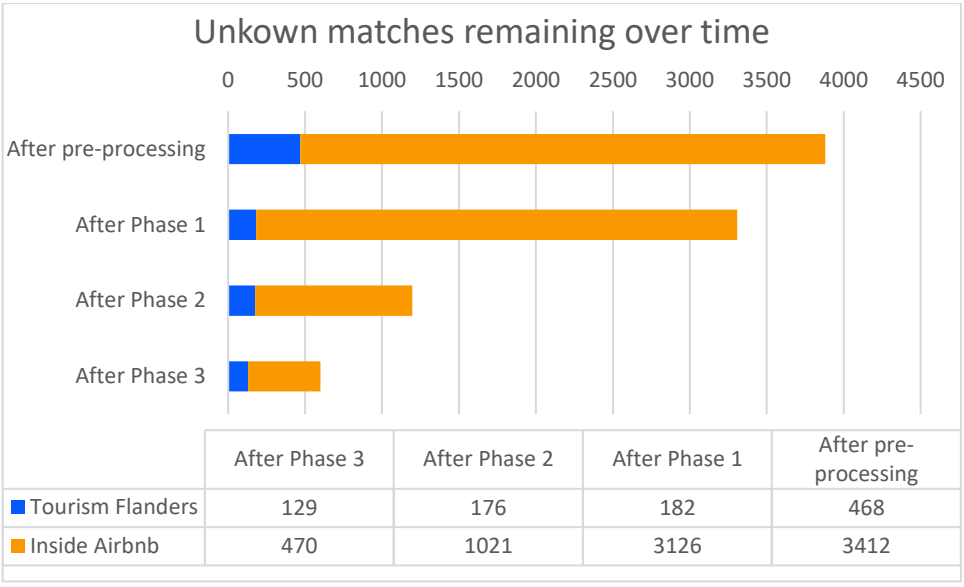


Fig. 9. Capacity check phase: unknown entries.

Fig. 10 presents a bar chart illustrating how both datasets were reduced at each phase, allowing for a clear comparison of data attrition between Tourism Flanders and Inside Airbnb.



	After Phase 3	After Phase 2	After Phase 1	After pre-processing
Tourism Flanders	129	176	182	468
Inside Airbnb	470	1021	3126	3412

Fig. 10. Illustrating the reduction in number of unknown matches in each phase.

5.2 Missing Locations

Across the first three phases of data matching, 2,656 locations were identified as missing from the Tourism Flanders dataset. During the fourth phase 362 additional locations were estimated as missed locations, bringing the total to 3,018 missed locations. This results in an approximation rate of 84%. A summary of these values is provided in Table 1.

Table 1. Missing locations.

Locations missed	Result (rounded to 1 unit)
Confirmed	2,656
Approximation ratio	84%
Approximate	362
Total	3,018

During the testing phase, 286 confirmed matches from the indexing phase were evaluated. Of these, 35 were excluded by the geospatial matching process, resulting in a recall rate of 87.76%. The remaining 251 were correctly included by the geospatial match. A summary of these results is presented in Table 2.

Table 2. Geospatial match recall.

Testing Geospatial match	Result (rounded to 1 unit)
Confirmed matches	286
incorrectly excluded	35
correctly included	251
Recall of geospatial match	87.76%

5.3 Capacity

Using the 3,018 confirmed missed locations, the missed capacity was calculated by summing the values of the accommodates variable, resulting in a confirmed missed capacity of 9,935. Since the specific accommodations for the 362 approximate entries are unknown, their capacity was estimated by multiplying the average accommodates value of the confirmed missed entries by 362. This resulted in an approximate missed capacity of 1,353. A summary of these values is provided in Table 3.

Table 3. Capacity.

Capacity	Result (rounded to 1 unit)
Original	533
Confirmed missed	9,935
Approximate missed	1,353
Total	11,821

5.4 Overnight Stays

By summing the original capacity (533), the confirmed missed capacity (9,935), and the approximate missed capacity (1,353), the total estimated capacity resulted in 11,821. Dividing the original capacity by the total estimated capacity yields a correction factor of approximately 22.18. The original statistics for overnight stay for each room type were provided as follows: 742,274 for GENERIC_ROOMS, 61,530 for BED_AND_BREAKFAST and 813,388 for HOLIDAY_COTTAGE. Since the pre-processed dataset did not have all the entries for these room types, each overnight stay statistic was adjusted based on the proportion of each room type available in the dataset relative to the

unprocessed totals. These values are provided in Table 4. The combined original total for the categories was 382,059 overnight stays. After applying the correction factor, the adjusted total number of overnight stays in Airbnb accommodations across Flanders is estimated at 8,481,695.

Table 4. Overnight stay.

Factor	22.17831
Overnight stay	Result (rounded to 1 unit)
Adapted BED_AND_BREAKFAST	9,323
Adapted GENERIC_ROOMS	274,277
Adapted HOLIDAY_COTTAGE	98,459
Total (Original statistic)	382,059

Adapted overnight stay statistic	8,481,695

6 Discussion

The research question posed by this paper is: How can data matching techniques enhance the real-world validity and reliability of tourism statistics for the Flanders region? This research proved that through data matching techniques the tourism statistics of Flanders can be improved in both reliability and real-world validity. This is shown in the results by how far the statistics from Tourism Flanders deviate from reality and highlight their lack of reliability. The discrepancy was noteworthy throughout the various parts of our results.

The first phase in the data matching process emphasises the importance of correct and clean data when faced with this kind of problem. For the Tourism Flanders dataset, it reduced the number of unknown entries by 61.11%. If all the links had been in a correct format, as opposed to Tourism Flanders' incomplete links this result would have been 100%. This would make the problem of finding all the missing locations more feasible and a case of pre-processing the data sets and performing a simple match. The second phase shows the importance of the geospatial coordinates in the dataset but also highlights again the issue with the inaccurate data. Despite its effectiveness, however, the approach was constrained by the set 172 metre range, which limited the precision of matches. A smaller range could have confirmed a higher rate of missed locations by Tourism Flanders but was infeasible due to limitations in the dataset.

It is important to take this error in location data into account when handling our data to make sure that the results correctly represent the real-life validity of the data.

The data matching results regarding missing locations also illustrate a significant discrepancy between Tourism Flanders' original count of Airbnbs and the count provided by Inside Airbnb. Specifically, Tourism Flanders' Airbnb listings accounted for only 13.43% of the total count presented by Inside Airbnb. To further assess the geospatial method, the paper tested the recall of the geospatial match which yielded a result of 87.76%. While a higher recall would have been preferred, this would have required increasing the 172-metre range. Although this would reduce the likelihood of excluding true matches, thereby increasing the measured accuracy, it would also include more false matches. As a result, the total effectiveness and real-world validity of the matching process would decrease, since a broader range may not accurately reflect actual property locations.

Another aspect that was aimed to find is the "True" value of missed locations. Although the result does provide a closer look at the reality of the location data, a portion of it is based on approximations which was due to the lack of matching descriptive variables left after the previous stages.

The original capacity calculated from the Tourism Flanders data represented 4.51% of the true value according to our results. This means that the calculated value can aid Tourism Flanders to better understand the capacity that Airbnb encompasses in the region as their previous records missed 95% of the calculated capacity. This is also a case where not having to use an approximation would have been preferred. The fact that Inside Airbnb contained the "accommodates" as a matching variable for the "max_capacity" in the Tourism Flanders dataset, however, speaks to accuracy of the results. The capacity missed based on the matches, is calculated in part with a matching variable, in contrast to if this value had to be entirely estimated. Following on from the missed capacity and using it to create a factor, the analyses revealed an estimated difference of 8,099,636 overnight stays com-pared to the original statistic provided by Tourism Flanders. These calculations should be more accurate than the other studies explored in the literature review due to the original statistic only needing to be adjusted instead of being entirely estimated.

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Overall, the discrepancies found between the two sources of statistics indicate that STRs have not been well tracked using existing methods. It demonstrates the relevance of studies like this for getting an idea of the adjustment

that needs to be made and to show just how far regulation would need to move forward before an accurate statistic is obtained. Another achievement of this study was automation. Through various techniques and data matching processes the study was able to locate a confirmed 3,018 missing locations with 11.99% being approximations. This was accomplished in a fraction of the time that it would have taken to perform by hand, a method that will be passed along to Tourism Flanders.

This paper is not just relevant to the specific case study or Flanders. It presents an adaptable methodology with key examples of techniques and methods to tackle the task of data matching across two very dissimilar data sets. An advantage that is ever more present in studies concerning Airbnb, as it uses the Inside Airbnb dataset which is commonly utilised due to its open-source nature. It can serve as a guideline to other users and government tourism agencies in how to adapt and apply this data.

As noted earlier, identifying missing locations is crucial for effective market regulation. Beyond accomplishing this, the paper provides guidelines on extracting valuable information from the process. Missing locations provide useful regulation insights, but the other touristic data serves further purposes as well. By creating a method for estimating missed capacity and overnight stays, this study provides Flanders with a more accurate estimate of visitor influx. These tourism statistics are important to understand and are utilised in many ways. For example, it can elaborate on the size of a countries GDP. The World Travel & Tourism Council states that “In 2024, Travel & Tourism's contribution to global GDP totalled US$ 10.9 trillion” (Travel & Tourism Economic Impact Research (EIR), 2025), emphasising the importance of tourism statistics.

7 Limitations and Avenues for Future Research

This study faces several limitations, related to data quality and the methodology used in the final matching phase. There are three key limitations related to the data itself, many of which reflect broader issues raised in the literature review. First, some individuals may register their accommodations under just a lodging as opposed to defining it as an Airbnb via the website variable. Second, the STR market is inherently volatile, with listings frequently being added and removed. Key variables, such as capacity, may also be outdated or inconsistent across time, making it difficult to track changes or verify data accuracy. These challenges were also noted by Prentice & Pawlicz (2023) and Alsudais (2021). Lastly, the use of scraped data from platforms like Inside Airbnb introduces uncertainty in the verification of listing authenticity. Listings may be mislinked or misrepresented, and essential variables such as reviews, categories, or addresses may be inaccurate or incomplete (Alsudais, 2021; Ferreri & Sanyal,

2018). These concerns about data accuracy, completeness and reproducibility, discussed extensively in prior research also apply to this study.

Another limitation, beyond the issues related to data quality, arises from the fourth phase of data matching and stems from two main factors. First, by estimating missed entries rather than identifying exact pairs, additional approximations were necessary when calculating missed capacity and overnight stays. Second, the lack of testability associated with this approximation method makes the accuracy of these calculations uncertain. An alternate method was considered using a machine learning model for the approximation phase. The research team hypothesised, however, that testing and training a machine learning algorithm would be unfeasible with the data used within this study. This was because the lack of matching descriptive variables after the first three phases and data quality would not provide a good enough base to train an algorithm. This opens the door for future research to confirm this hypothesis or to use a different dataset. Further research could also explore the scalability of the proposed methodology. Testing its applications across other tourism regions or even different STR platforms, independently of any integration of machine learning. Another avenue for future research would be looking into other methods than Haversine for calculating the geospatial distances based on coordinates or testing different thresholds for this papers' objectives.

8 Conclusion

This paper attempted to answer the research question through the three objectives outlined by Tourism Flanders. The first objective was to find the accommodations Tourism Flanders missed. The second objective was to use these missing accommodations to calculate the capacity missed. The last objective was to generate a factor to adjust Tourism Flanders overnight stays statistic. Consistent with previous literature, it was necessary to overcome the limitations and data inconsistencies associated with STR databases specifically with Inside Airbnb to achieve these objectives. This paper handled these challenges by accounting for inaccuracies in datapoints and addressed the coordinate inaccuracy issues associated with geospatial data across datasets, which has not previously been accounted for throughout the literature in this study. These inaccuracies were directly addressed in the methodology.

The methodology began with a thorough pre-processing of both the Inside Airbnb and Tourism Flanders datasets which proved to be important as it reduced each dataset drastically. The data matching method was divided into four phases. The first phase used an indexing technique and matched the unique listing IDs from Inside Airbnb to the web addresses from Tourism Flanders. Phase two employed geospatial matching utilising the Haversine formula to compare coordinate data within a defined 172-meter threshold that

accounted for known inaccuracies. Phase Three compared the accommodation capacities present in both datasets, to further refine potential matches and identify non-matches. The fourth and final phase, derived the approximation ratio from the confirmed matches and non-matches of earlier stages, to estimate the total number of Airbnb listings likely missed by Tourism Flanders. These techniques minimise manual intervention and enhance efficiency. The matching process was designed for maximal automation. The research achieved this by leveraging matching methodologies appropriate for the specific variables and datasets involved.

These objectives lead the study into further understanding the effects of using these data matching techniques and answering the research question: How can data matching techniques enhance the real-world validity and reliability of tourism statistics for the Flanders region? Our results suggest that our data matching techniques illustrate the sheer scale of missed Tourism information and the wealth of real-life data that could have been found had these techniques been applied previously. Through removing inaccurate entries and validating the accuracy of our matches the reliability of these results increased.

This structured approach not only facilitated more efficient data matching but also demonstrated a pathway to potentially reduce the considerable time investment typically required for such processes. Further research could focus on three separate avenues. Firstly, exploring the integration of machine learning for dynamic correction factor estimation. Secondly investigate the scalability of the methodology to other regions and platforms. Thirdly, optimising the geospatial distance measure for matching between this studies' datasets.

In conclusion, by systematically identifying and providing a means to account for previously untracked STR activity, this thesis contributes to enhancing the validity and reliability of tourism statistics. It equips stakeholders like Tourism Flanders with a practical methodology and a quantifiable understanding of the STR market's true scale, thereby fostering more informed strategic planning, effective resource allocation, and the development of sustainable tourism policies in an increasingly digital and shared economy.

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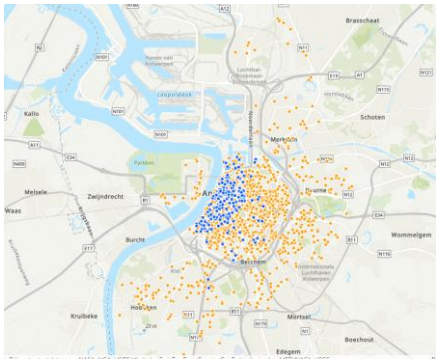
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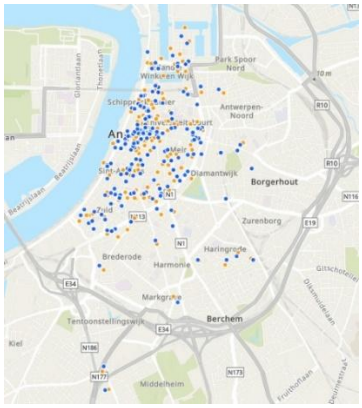
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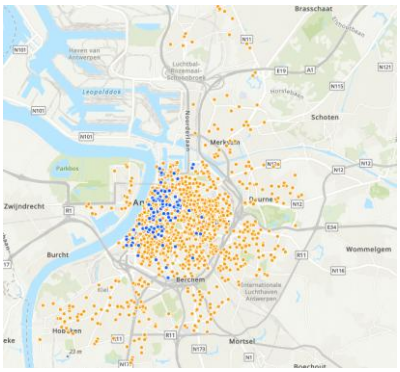
Appendix 1



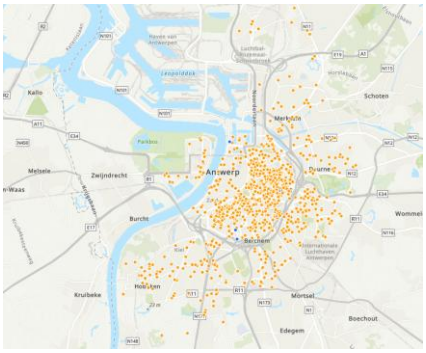
Antwerp original pre-processed data entries from Inside Airbnb and Tourism Flanders.



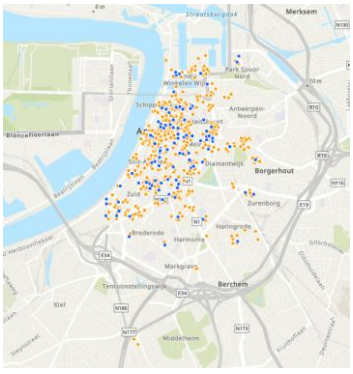
Antwerp Indexing Phase: Visualisation of the matched entries.



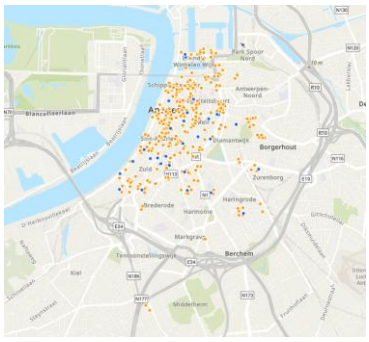
Antwerp Indexing phase: Visualisation of the unmatched and unknown entries.



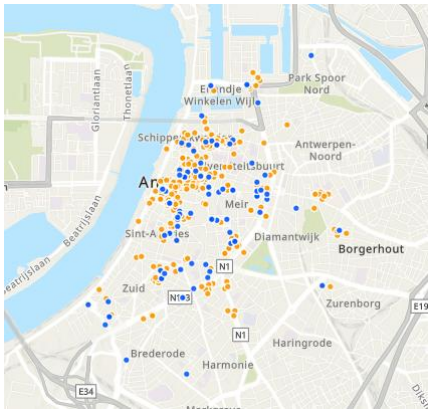
Antwerp Geospatial matching phase: Visualisation of the confirmed non matches.



Antwerp Geospatial matching phase: Visualisation of Unknown data points.



Antwerp Capacity check phase: confirmed non-matches.



Antwerp Capacity check phase: unknown entries**.**

Appendix 2

Link to code used in the paper as a github link: thijsdb3/master_thesis

List of Figures

Fig. 1. Pre-processing BPMN, outlining the general pre-process, including filter, clean, fixing web addresses, checking for distinct active listings and merge data. ..... 1**Error! Bookmark not defined.**

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Fig. 1. Pre-processing BPMN, outlining the general pre-process, including filter, clean, fixing web addresses, checking for distinct active listings and merge data. 17

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